

Pandas: Handling Missing Values in CSV Files

1. Import Pandas

```
import pandas as pd
```

Notes

- `pandas` is a Python library used for data analysis and manipulation.
 - `pd` is an alias for pandas.
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2. Read Data from CSV File

```
df = pd.read_csv("data.csv")
```

Notes

- `read_csv()` reads data from a CSV file.
 - The data is stored in a DataFrame called `df`.
 - A DataFrame is a table consisting of rows and columns.
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3. Print DataFrame

```
print(df)
```

Notes

- Displays the entire contents of the CSV file.
 - Useful for verifying that the data has been loaded correctly.
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Handling Missing Values

Missing values in pandas are represented as `NaN` (Not a Number).

Example:

Name	Age
Ali	20

Name	Age
Sara	NaN
Ahmed	22

4. Replace Missing Values with Mean

```
df["Age"] = df["Age"].fillna(df["Age"].mean())  
print(df)
```

Explanation

- `mean()` calculates the average value of the column.
- `fillna()` replaces missing values with the calculated mean.

Example

Original Data:

```
20  
NaN  
22
```

Mean:

```
(20 + 22) / 2 = 21
```

Result:

```
20  
21  
22
```

Use Case

- Best for numerical data without extreme outliers.

5. Replace Missing Values with Median

```
df["Age"] = df["Age"].fillna(df["Age"].median())  
print(df)
```

Explanation

- `median()` returns the middle value after sorting.
- Missing values are replaced with the median.

Example

Data:

```
20  
22  
30
```

Median:

```
22
```

Result:

```
20  
22  
22  
30
```

Use Case

- Useful when data contains outliers.
- More robust than mean.

6. Replace Missing Values with Mode

```
df["Age"] = df["Age"].fillna(df["Age"].mode()[0])  
print(df)
```

Explanation

- `mode()` returns the most frequently occurring value.
- `[0]` selects the first mode value.

Example

Data:

20
20
22
30

Mode:

20

Result:

20
20
20
22
30

Use Case

- Commonly used for categorical data.
- Can also be used for numerical data with repeated values.

Summary Table

Method	Function	Purpose
Mean	<code>mean()</code>	Replace missing values with average
Median	<code>median()</code>	Replace missing values with middle value
Mode	<code>mode()</code>	Replace missing values with most frequent value

Original Code

```
import pandas as pd

# Read CSV file
df = pd.read_csv("data.csv")

print("#####")

# Replacing missing values with mean
# df["Age"] = df["Age"].fillna(df["Age"].mean())
```

```
# print(df)

print("#####")

# Replacing missing values with median
# df["Age"] = df["Age"].fillna(df["Age"].median())
# print(df)

print("#####")

# Replacing missing values with mode
# df["Age"] = df["Age"].fillna(df["Age"].mode()[0])
# print(df)
```